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The Journal of Sports Medicine and Physical Fitness 2024 Sep 25
DOI: 10.23736/S0022-4707.24.16199-3

REVIEW

SPORT INJURIES AND REHABILITATION

Is chronic ankle instability associated with contractile thickness of gluteus medius and gluteus maximus during functional movement and exercise? A systematic review and meta-analysis

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ABSTRACT

INTRODUCTION: The gluteus medius (GMED) and gluteus maximus (GMAX) play a crucial role in postural control, and postural control is impaired in individuals with chronic ankle instability (CAI). However, the association between CAI and the recruitment of these muscles remains unclear. The purpose of this study was to explore the contractile thickness of GMED and GMAX during functional movements in individuals with CAI compared to healthy controls.
EVIDENCE ACQUISITION: A systematic search was conducted in six databases (PubMed, Embase, Cochrane Library, Web of Science, EBSCO, and PEDro). Included studies involved the contractile thickness of GMED and/or GMAX comparing CAI with non-CAI. The extracted data were subjected to meta-analysis for observing the differences between the two. The correlation and difference in contraction between GMED and GMAX were analyzed using the Pearson Correlation Coefficient (PCC) and *t*-test, respectively.
EVIDENCE SYNTHESIS: Six studies with 176 participants were found. Contractile thickness measured by ultrasound showed GMED thickness change to be lower in CAI than in controls during functional movements and exercise (WMD: -0.08; 95% CI: -0.11, -0.04; P<0.00001). There was no significant difference between the two groups with respect to contraction of GMAX (WMD: 0.02; 95% CI: -0.01, 0.05; P=0.25). The PCC and P value (*t*-test) between the ratio of contractile thickness of GMED and GMAX in CAI were 0.397 and 0.029 respectively, indicating activity differences.
CONCLUSIONS: CAI may be associated with weaker GMED recruitment during functional movements and exercise, but the activation of GMAX in CAI may be unaffected.

(*Cite this article as:* Luan L, Witchalls J, Ganderton G, Adams R, El-Ansary D, Han J. Is chronic ankle instability associated with contractile thickness of gluteus medius and gluteus maximus during functional movement and exercise? A systematic review and meta-analysis. J Sports Med Phys Fitness 2024 Sep 25. DOI: 10.23736/S0022-4707.24.16199-3)

KEY WORDS: Ankle joint; Joint instability; Muscle contraction; Systematic review.

Introduction

Chronic ankle instability (CAI) is a condition that is characterized by frequent episodes of "giving way" and feelings of having unstable ankles, and it often results from ankle injuries, especially in physically active individuals and athletes.[1] Individuals with CAI may suffer from recurrent ankle sprain, and are also likely to have

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poor posture control, which can significantly affect their quality of life and sport performance.[2-4]

In terms of posture control, the gluteus medius (GMED) and gluteus maximus (GMAX) are thought to play an important role in stabilizing the pelvis and controlling the femur in functional movements,[5] particularly in single leg loading,[6] based on their biomechanical characteristics.[2, 7, 8] The GMED originates from the outer surface of the ilium and the gluteal aponeurosis and inserts onto the lateral surface of the greater trochanter of the femur, and has three functionally independent segments.[9, 10] Its anterior portion contributes to hip flexion and internal rotation. Its posterior portion participates in hip extension and external rotation.[11, 12] Arguably, good postural control in physical activities would be difficult to achieve if the GMED is not recruited sufficiently.[13, 14]

Similarly, the GMAX plays a substantial role in posture control and physical activities, although its specific biomechanical characteristics and contributions differ from GMED. The GMAX originates from the posterior ilium, sacrum, and coccyx, and inserts into the iliotibial tract and gluteal tuberosity of the femur.[9] Its primary actions also involve hip extension, external rotation, and abduction, playing a key role in activities such as standing up from a sitting position, climbing stairs, and running.[10, 12] Abundant recruitment of GMAX is essential for providing stability during movements that require extension of the hip, pointing to its importance in various daily and athletic tasks.[15, 16] In summary, any reduction of GMED and GMAX recruitment may result in impaired postural control.

Given that patients with CAI have shown impaired postural control, one possible mechanism in this is that the recruitment capacity of the GMED and/or GMAX are reduced.[6, 17] This proposition is supported by previous studies reporting that individuals with CAI may not only exhibit deficiencies in ankle strength but also experience a decline in the recruitment capacity of the gluteal muscles,[4, 18] and evidence suggesting that postural control could be enhanced in CAI individuals through gluteal muscles strengthening.[11, 17, 19] Thus, investigating the recruitment capacity of the GMED and GMAX is essential for studying postural control in CAI.

To reflect the recruitment of these muscles, electromyography (EMG) is commonly used, and previous studies have extensively employed this method.[7, 12, 20] Another effective method for evaluating muscle recruitment is to observe contractile thickness, which is generally quantified using ultrasound imaging, and is regarded as safe and a reliable measure of muscle thickness.[3, 5] This approach may also provide information regarding the degree of muscle recruitment because during muscle activation, nerve impulses stimulate muscle contraction, causing muscle fibers to shorten, which results in visible and measurable changes in muscle thickness.[21] Additionally, during muscle recruitment, blood flows to the activated muscle area to provide oxygen and nutrients, which may lead to congestion and swelling in the muscle region, thereby affecting muscle thickness.[22] Changes in the water content of muscle tissue may occur during muscle contraction or congestion, further influencing muscle thickness.[23] Therefore, the muscle contractile thickness can partly reflect muscle recruitment.

Currently, there have been some studies that have used ultrasound to investigate the contractile thickness of GMED and/or GMAX during physical activities, comparing individuals with CAI to individuals without CAI. However, the reported results from these studies are not consistent. Some trials have found that CAI is significantly weaker than non-CAI, while others have found CAI to be significantly stronger; also, some trials have not observed significant differences between the two.[3, 5, 15, 16] In brief, the research on this subject is still inconclusive at present. Additionally, it remains unclear whether both the GMED and GMAX in CAI are abnormal, or if only one of them is affected, as the correlation and difference between the recruitment capacity of GMED and GMAX in CAI have not yet been investigated.[7] Therefore, the aim of this study was to elucidate, through a systematic review, differences in the contractile thickness of GMED and GMAX between individuals with and without CAI, as well as the relations (correlation and difference) between the recruitment of the GMED and GMAX.

Evidence acquisition

Data sources and searches

This study was performed in accordance with the PRISMA guidelines,[24] and registered in the International Prospective Register of Systematic Reviews (PROSPERO) database (registration number: CRD42024497002).

A systematic search was conducted using the following electronic databases: PubMed, Embase, Cochrane Library, Web of Science, EBSCO and PEDro, from inception to 15 January 2024. There was no restriction on publication type and language. The search strategy used the terms from the three areas: (1) ankle instability ("ankle instability," "instability, ankle," "instable ankle," "unstable ankle"), (2) gluteal muscles ("glutes*," "gluteal muscle*," "gluteus

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muscle*," "gluteus medius*," "gluteus medius muscle*," "gluteus medium*," "gluteus medium muscle*," "gluteus maximus*," "buttock* muscle*"), and (3) muscle size ("thickness*," "size*," "cross sectional area*," "CSA*," "ultrasound," "ultrason*," "echo*," "radio*," "diagnos*," "*photo*," "*graphy*," "*imag*," "functional activity ratio*," "functional activation ratio*," "FAR*"); at a location in the title and/or abstract and/or keywords, each term in the area was merged with the operation of "or," and an "and" operator was used for combining between areas.

Study selection

Two reviewers (LL and JH) independently assessed all obtained studies. After removing duplicates, the titles and abstracts of the studies were screened. Full texts of the potential studies were reviewed to obtain eligible trials.

The inclusion criteria were as follows: 1) participants: individuals with CAI; 2) intervention: physical activities; 3) comparator: healthy population or copers (individuals who have experienced one significant ankle sprain at least 12 months before study collection without any symptoms of CAI); 4) outcome: the contractile thickness of GMED and/or GMAX; 5) study design: observational study.

The exclusion criteria were: 1) studies conducted with animals, cadavers, simulators, prostheses, or in vitro; 2) trials investigating certain shoes or footwear, such as those involving foot orthoses, ankle braces/devices, and kinesiology taping; 3) study design/publication type: meeting abstract, trial registry, detection, survey, case report, and review.

Quality assessment

The literature quality assessment was conducted independently by two assessors (LL and JH) using the Newcastle-Ottawa Quality Scale (NOS).[25, 26] The scores for each study were achieved by the two assessors reaching consensus through discussion, and any disagreements were resolved by a third assessor (RA). Output from the NOS contains 9 items with a total score out of 9. Each item is marked "*" (1 point) or null (0 point); a score above 7 indicates high quality, a score from 4 to 6 presents moderate quality, and scores of less than 3 reflect poor quality.[27, 28]

In addition, the GRADE system rating (software: GRADE Profiler Version 3.6) was used to assess the quality of evidence, which provides explicit criteria for downgrading the quality of evidence that include risk of bias, inconsistency, indirectness, and imprecision, and extra considerations for upgrading the quality of evidence, such as effect, confounding, and dose. For the observational studies, the quality of evidence is related to the combined results of the above items. Where there were disagreements in rating, consensus was reached through discussion.

Data extraction and outcome measures

Ultrasound was used to determine muscle thickness, and the reported differences in contractile thickness between those with CAI and those in control groups were analyzed. A common measurement is the Functional Activity Ratio (FAR), which is calculated as the ratio of muscle thickness during activity to muscle thickness at rest, and serves as a valuable indicator for analyzing muscle contractile thickness during functional movements; lower/higher FAR indicates less/more contractile thickness.[5, 16] Although ultrasound imaging can generate various muscle size parameters such as thickness, cross-sectional area, and quality (echogenicity) of muscle fibers,[5, 15] the FAR is the primary reference metric for outcome measurement in this study, as it goes beyond pure size assessment, offering a more comprehensive understanding of muscle recruitment during different activities. This versatile metric aids in evaluating function and dynamic muscle performance, providing nuanced insights into the impact of muscle thickness variations on functional activities.

Data synthesis and statistical analysis

In terms of the study on the differences in the ratio of contractile thickness between CAI and non-CAI, RevMan 5.3 software was employed to conduct a meta-analysis of FARs from various studies. The assessment revealed that studies with I-square values exceeding 50% exhibited high heterogeneity, while those below 50% indicated better homogeneity. A P value less than 0.05 was considered statistically significant, emphasizing meaningful differences. In the context of walking movements, FAR values were averaged across the entire gait cycle. Some studies presented graphical data only, without explicit numerical values. In such instances, a meticulous approach was undertaken, involving graphical scaling and coordinate axis value extraction, to convert graphical representations into precise numerical data.

After obtaining data based on FAR calculations, the relations in the ratio of contractile thickness between GMED and GMAX were investigated. To begin with, the correlation between the FAR values of GMED and GMAX was analyzed using the Pearson Correlation Coefficient (PCC) (strong positive correlation: PCC>0.7, moderate positive correlation: 0.5<PCC≤0.7, weak positive correlation: 0.3<PCC≤0.5, no significant correlation:

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|PCC|≤0.3), and the higher/lower correlation indicates a stronger/poorer consistency of the ratio of contractile thickness. Additionally, the difference between the FAR values of GMED and GMAX was analyzed using the *t*-test, including two indicators: the P value, reporting the statistically difference (significance level: 0.01≤P<0.05, highly significant: P<0.01), and the Cohen's-D (d), reflecting the degree of difference (very large effect: d>0.8, large effect: 0.5<d≤0.8, medium effect: 0.2<d≤0.5, small effect: d≤0.2). While the calculation of these values requires a large sample size to derive relatively accurate and valid conclusions, this study is merely aimed at exploring trends, regardless of the sample size.

Evidence synthesis

Literature search and screening

The flow diagram of the detailed selection process, following the PRISMA guidelines, is shown in Figure 1. A total of 201 relevant articles were obtained from the 6 databases. After the removal of duplicates, the title and abstract of 133 records were screened, and full texts of 75 studies were reviewed. Finally, after screening whether the study subjects involve CAI and whether the study observations include measurements of the thickness changes of the gluteal muscles during physical activities, 6 articles (176 participants) were judged to be eligible for inclusion,[3, 5, 15, 16, 29, 30] and 4 of these were included in the meta-analysis.[3, 5, 15, 16]

Quality assessment of the included studies

The NOS stars scored for each study are given in Table I.[3, 5, 15, 16, 29, 30] Among the six studies included, four were found to be of high quality, and all scored 9 NOS stars.[3, 5, 15, 16] The remaining two studies consisted of one that did not follow an explicit controlled design,[29] and the other had a control group composed of individuals with CAI,[30] precluding their inclusion in the NOS assessment. As a result, these two studies were not assigned scores in the quality evaluation process.

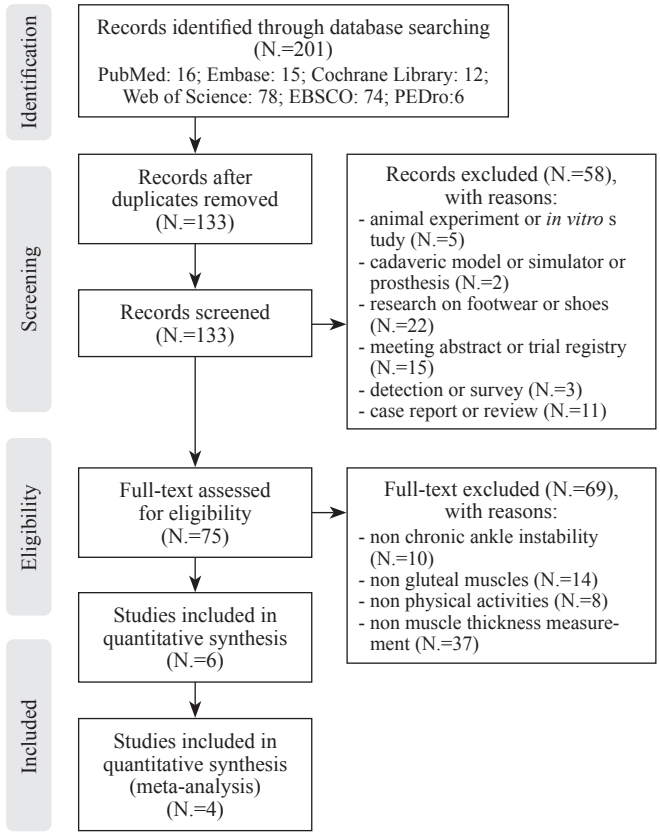

Figure 1.—Flow chart of study selection.

Characteristics of the eligible studies

Participant characteristics are detailed in Supplementary Digital Material 1, Supplementary Table I,[3, 5, 15, 16, 29, 30] with the included studies spanning the years 2019 to 2022. All participants in these studies met the CAI inclusion criteria

TABLE I.—*Newcastle-Ottawa scale stars of included studies.*[3, 5, 15, 16, 29, 30]

Study	Selection				Comparability		Exposure/Outcome			Total stars
	S1	S2	S3	S4	C1	C2	E1	E2	E3	
Koldenhoven[3]	*	*	*	*	*	*	*	*	*	9
DeJong[5]	*	*	*	*	*	*	*	*	*	9
DeJong[15]	*	*	*	*	*	*	*	*	*	9
DeJong[16]	*	*	*	*	*	*	*	*	*	9
DeJong[29]	-	-	-	-	-	-	-	-	-	-
DeJong[30]	-	-	-	-	-	-	-	-	-	-

S1: definition of cases; S2: representativeness of the cases; S3: selection of controls; S4: adequate control definition; C1: comparability of cases; C2: study controls for the basis of the analysis; E1: ascertainment of the exposure; E2: ascertainment of the same method used for cases and controls; E3: non-response rate.

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set by the International Ankle Consortium. Notably, two studies included copers in the control group.[3, 15] The investigated motor tasks in these studies primarily encompassed walking, balance assessments, and resistance training.

Concise summaries of the included studies are presented in Supplementary Digital Material 2, Supplementary Table II.[3, 5, 15, 16, 29, 30] The assessment of contractile thickness of GMED and GMAX in all studies involved the use of ultrasound. While variations existed across studies in terms of ultrasound equipment, testing parameters, procedures, and data acquisition and processing, the overall methodologies were largely consistent. Additionally, each study comprehensively reported on the conditions of both the GMED and GMAX.

Meta-analysis of GMED

When comparing individuals with CAI to the control group, the former exhibited significantly lower FARs in the GMED (WMD: -0.08; 95% CI: -0.11, -0.04; P<0.00001). However, a notable heterogeneity was observed with an I^2 value of 74%, indicating substantial heterogeneity among the studies.[3, 5, 15, 16]

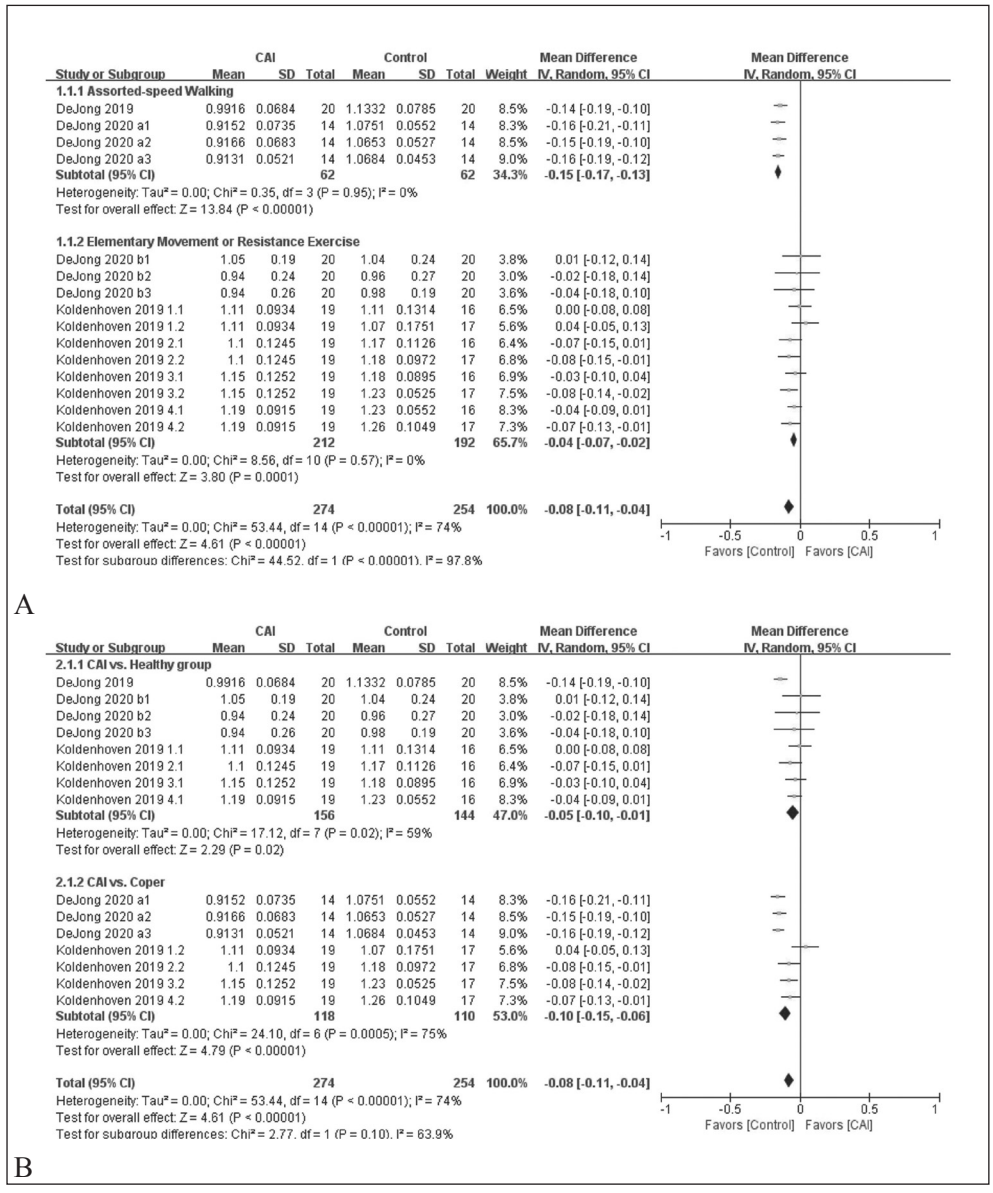

Figure 2.—Comparison of Functional Activity Ratios (FARs) in the Gluteus Medius: (A) Forest plots depicting CAI and control groups across various tasks; (B) Forest plots comparing CAI with different control groups.[3, 5, 15, 16]

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Further stratification based on the primary characteristics of the motor tasks revealed two subgroups: 1) the assorted-speed walking subgroup, where CAI GMED FARs were significantly lower than the control (WMD: -0.15; 95% CI: -0.17, -0.13; P<0.00001, I^2=0%);[5, 15] 2) the elementary movement or resistance exercise subgroup, where CAI GMED FARs were also significantly lower than the control (WMD: -0.04; 95% CI: -0.07, -0.02; P=0.0001, I^2=0%).[3, 16] Forest plots detailing the specifics of these meta-analyses are presented in Figure 2A.[3, 5, 15, 16]

Additionally, considering the characteristics of the control groups, two subgroups were identified: 1) CAI *vs.* Healthy group, where CAI GMED FARs were significantly lower (WMD: -0.05; 95% CI: -0.10, -0.01; P=0.02, I^2=59%);[3, 5, 16] 2) CAI *vs*. Coper, where CAI GMED FARs were also significantly lower (WMD: -0.10; 95% CI: -0.15, -0.06; p<0.00001, I^2=75%).[3, 15] Forest plots detailing the specifics of these meta-analyses are presented in Figure 2B.[3, 5, 15, 16]

The GRADE rating for the above meta-analysis results of the GMED is reported in section A of Table II. Some assessment items were downgraded due to heterogeneity and small sample sizes; however, most of the meta-analysis results showed a large and significant effect size, leading to an upgrade in scores. Overall, the quality of evidence for the GMED overview is rated as moderate.

Meta-analysis of GMAX

When comparing the FARs of the GMAX between individuals with CAI and the control group, no significant difference was observed (WMD: 0.02; 95% CI: -0.01, 0.05; P=0.25). However, the relatively high I^2 value (71%) indicated considerable heterogeneity among the studies.[3, 5, 15, 16]

Similarly, subgroup analysis based on the primary characteristics of the motor tasks resulted in two subgroups: 1) the assorted-speed walking subgroup, where CAI GMAX FARs showed no significant difference compared to the control group (WMD: 0.07; 95% CI: -0.01, 0.16; P=0.08, I^2=86%);[5, 15] 2) the elementary movement or resistance exercise subgroup, where CAI GMAX FARs also exhibited no significant difference compared to the control group (WMD: -0.00; 95% CI: -0.03, 0.02; P=0.84, I^2=52%).[3, 16] Detailed forest plots illustrating these meta-analyses can be found in Figure 3A.[3, 5, 15, 16]

TABLE II.—*Rating of evidence quality of meta-analysis results using the GRADE tool.*

GRADE System Assessment							
Number of trials (Participants)	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Upgrade consideration (effect)	Quality of evidence
(A) Gluteus Medius							
Gluteus medius - overview							
15 (528)	No serious	Serious*	No serious	No serious	Undetected	Very large**	Moderate
Gluteus medius - assorted-speed walking							
4 (124)	No serious	No serious	No serious	Serious***	Undetected	Very large**	High
Gluteus medius - elementary movement or resistance exercise							
11 (404)	No serious	No serious	No serious	No serious	Undetected	Very large**	High
Gluteus medius - CAI *vs*. healthy group							
8 (300)	No serious	Serious*	No serious	No serious	Undetected	Large****	Low
Gluteus medius - CAI *vs*. Coper							
7 (228)	No serious	Serious*	No serious	No serious	Undetected	Very large**	Moderate
(B) Gluteus maximus							
Gluteus maximus - overview							
15 (528)	No serious	Serious*	No serious	No serious	Undetected	None	Very low
Gluteus maximus - assorted-speed walking							
4 (124)	No serious	Serious*	No serious	Serious***	Undetected	None	Very low
gluteus maximus - elementary movement or resistance exercise							
11 (404)	No serious	Serious*	No serious	No serious	Undetected	None	Very low
Gluteus maximus - CAI *vs*. healthy group							
8 (300)	No serious	No serious	No serious	No serious	Undetected	None	Low
Gluteus Maximus - CAI *vs*. Coper							
7 (228)	No serious	Serious*	No serious	No serious	Undetected	None	Very low

GRADE Working Group grades of evidence
High quality: Further research is very unlikely to change our confidence in the estimate of effect. Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate. Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate. Very low quality: We are very uncertain about the estimate. *Heterogeneity: I^2>50%; **test for overall effect: Z>2.58 (P<0.01); ***small sample size; ****test for overall effect: Z>1.96 (P<0.05).

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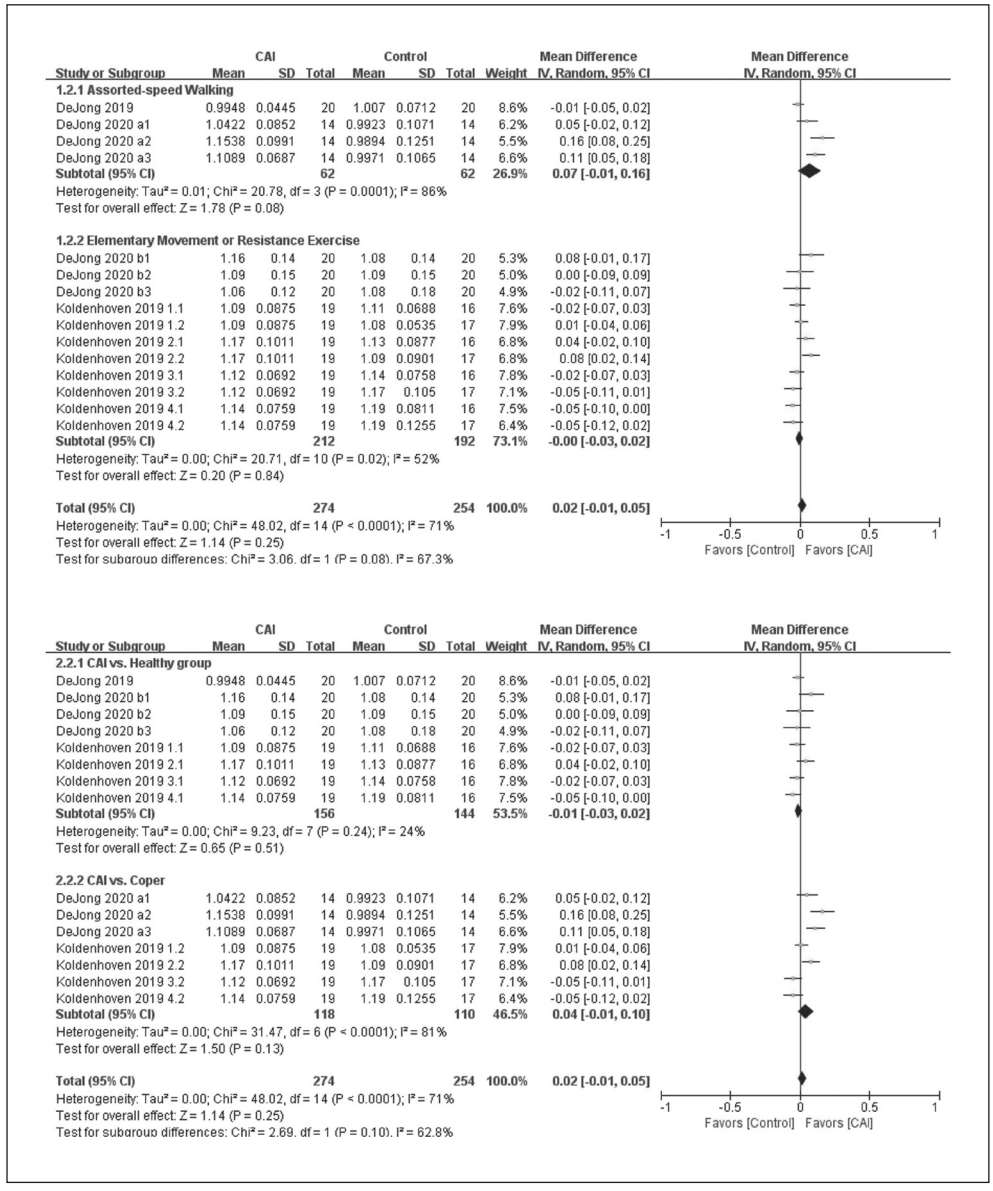

Figure 3.—Comparison of Functional Activity Ratios (FARs) in the Gluteus Maximus: (A) Forest plots depicting CAI and control groups across various tasks; (B) Forest plots comparing CAI with different control groups.[3, 5, 15, 16]

Subgroup analysis based on the characteristics of the control group resulted in two subgroups: 1) CAI *vs.* a Healthy group, where the FARs of the GMAX showed no significant difference between the two groups (WMD: -0.01; 95% CI: -0.03, 0.02; P=0.51, I^2=24%);[3, 5, 16] 2) CAI *vs.* Copers, where the FARs of the GMAX also exhibited no significant difference between the two groups (WMD: 0.04; 95% CI: -0.01, 0.10; P=0.13, I^2=81%).[3, 15] Detailed forest plots illustrating these meta-analyses can be found in Figure 3B.[3, 5, 15, 16]

The GRADE rating for the above meta-analysis results of the GMAX is reported in section B of Table II. Some assessment items were downgraded due to heterogeneity and small sample sizes. Additionally, since these meta-analysis results did not show a large and significant effect size, no upgrade in scores was awarded. Overall, compared to the results of the GMED studies, the evidence quality for the GMAX studies is generally lower due to the lack of clear effect trends.

Relation between GMED and GMAX

Table III provides an overview of the relations between FAR values for GMED and GMAX within various groups, employing the Pearson Correlation Coefficient (PCC) for

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TABLE III.—*Relation between FAR values for gluteus medius and gluteus maximus.*

Group	Correlation between the FAR values of GMED and GMAX	Difference (*t*-test) between the FAR values of GMED and GMAX	
	PCC	P value	Cohen's-D
CAI	0.397	0.029	-1.001
Control (Healthy group & Coper)	0.627	0.354	0.344
Healthy group	0.517	0.941	-0.038
Coper	0.922	0.191	0.740

FAR: Functional Activity Ratio (muscle thickness during activity / muscle thickness quiet); GMED: gluteus medius; GMAX: Gluteus Maximus; PCC: Pearson Correlation Coefficient.

correlation analysis, as well as the *t*-test (P value and Cohen's-D) for difference analysis. In the CAI group, a weak positive correlation is observed (PCC: 0.397), and a statistically significant difference (P=0.029) with a very large effect size (Cohen's-D: -1.001) is reported. Conversely, the Control group, comprising both the healthy group and coper subgroups, exhibits a moderate positive correlation (PCC: 0.627) with an insignificant difference (P=0.354) and a medium effect size (Cohen's-D: 0.344).

When considering the healthy group independently, a moderate positive correlation (PCC: 0.517) is found, yet without a significant difference (P=0.941) and with a small effect size (Cohen's-D: -0.038). In addition, the coper group demonstrates a strong positive correlation (PCC: 0.922) without a significant difference (P=0.191) and a large effect size (Cohen's-D: 0.740). These results highlight the varying degrees of correlation between GMED and GMAX in different groups, emphasizing levels of difference with both the statistical significance and effect size nuances.

Discussion

This review has identified that individuals with CAI exhibit lower GMED FAR values during assorted-speed walking and elementary movement or resistance exercise, when compared to controls. No significant differences were found for GMAX FAR values. Additionally, this study reveals a weaker correlation between GMED and GMAX in CAI, with statistically significant differences in their thickness changes during activity, presenting an inconsistent pattern of recruitment. In contrast, the control group demonstrates a stronger correlation between GMED and GMAX, with insignificant differences in the ratio of contractile thickness, indicating similar thickness changes during activity.

The contractile thickness of GMED and GMAX can be related to their recruitment capacity during physical activities,[31, 32] which play an important role in postural control.[21, 23, 30] It has been reported that when the muscle is activated and is working, blood flows into it, and more muscle fibers would be recruited, such that the muscle shape may be greater.[33] Increased thickness during recruitment of GMED and GMAX likely indicates that the individual has good neuromuscular control over GMED and GMAX.[32]

Building upon these concepts and considering current results indicating that the change of GMED thickness in individuals with CAI during physical activities is less compared to the control group, these findings suggest that the recruitment capacity of GMED may be weaker in CAI populations. However, the contractile thickness of GMAX in CAI appears comparable to the control group, with no significant difference in the change of GMAX thickness during activities, indicating that GMAX in CAI remains good recruitment capacity.

GMED in CAI

The meta-analysis results demonstrate that GMED FAR values in CAI are significantly lower than those of the control group, albeit with heterogeneity. When divided into subgroups based on the type of movement, such as assorted-speed walking and elementary movement or resistance exercise, both subgroups homogeneously show significantly lower FAR values in CAI compared to the control group (I^2=0%). Notably, the difference is more pronounced in the assorted-speed walking subgroup.

Elementary movement or resistance exercise involves deliberate actions, typically requiring participants to intentionally perform goal-oriented movements.[3, 16] In the context of the current study, hip abduction control is the focus. During such activities, the neuromuscular control around the CAI GMED might be more active due to focused attention but still inferior to the control group. Additionally, during assorted-speed walking, where the emphasis is on the direction of movement rather than specific muscle recruitment,[5, 15] the diminished focus on GMED might contribute to its reduced activity in individuals with CAI compared to the control group. From this, it can be observed that, regardless of the type of movement, the recruitment of GMED in individuals with CAI appears to be weaker than that in the control group, with unconscious movement patterns possibly exhibiting even greater deficits. Previous studies have shown that a weak recruitment capacity of GMED can cause hip instability and impair postural control.[7, 8, 22] Whether CAI causes GMED weakness, possibly through disuse, or having a weak GMED leads to CAI, remains to be clarified by future work.

When reclassified based on the characteristics of the

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control group into CAI vs. healthy group and CAI vs. coper subgroups, similar results are observed, with patients with CAI demonstrating recruitment deficits compared to both groups. However, there is high heterogeneity (I^2>50%) in these meta-analysis results. This raises the question as to whether, despite adhering to the inclusion criteria recommended by the International Ankle Consortium,[34] there are still significant differences among individuals with CAI. It is evident that, despite being categorized as CAI, different studies present inconsistent findings in their reports on electromyographic (EMG) signals, kinematics, and kinetics.[7, 8, 35, 36] One study even suggests the existence of six distinct movement control strategies within individuals with CAI.[37] This has been demonstrated in studies of other performance characteristics.[38] These disparities are likely a significant contributing factor to the observed high heterogeneity observed here. Consequently, there is an urgent need for a more precise definition of CAI in future research.

GMAX in CAI

Current meta-analysis results show no significant difference in FAR values between CAI and the control group for the GMAX. Subgroup analysis based on movement types and control group characteristics consistently supports this finding. The implication is that, during physical activities, individuals with CAI exhibit GMAX activation that is comparable to that of healthy individuals or copers.

The neuromuscular control of GMAX involves intricate coordination between motor neurons and muscle fibers, facilitating precise and efficient movements during various physical activities.[17] The dynamic recruitment of muscle fibers based on task demands underscores the adaptability of the GMAX to diverse movement patterns.[11, 19] When exploring differences in GMAX activity between individuals with CAI and the control group, it becomes evident that both cohorts exhibit similar recruitment patterns. This similarity can be attributed to the inherent nature of human motor control, where individuals develop consistent recruitment habits for the GMAX during activities such as walking, balancing, and resistance exercises.[39-41] Additionally, the commonality in GMAX recruitment highlights the significance of motor learning and muscle memory in shaping movement patterns.[42-44] Regardless of ankle stability status, individuals share habitual behaviors in recruiting the GMAX,[45] contributing to the observed minimal differences in GMAX activity between CAI and control groups. Finally, another explanation for the lack of difference in GMAX may be that it is more active than GMED in hip extension and this movement direction may be less compromised by the inversion risks experienced by participants with CAI. As a result, GMED is affected but GMAX stays activated.

In summary, the unique features of GMAX and its role in neuromuscular control contribute significantly to maintaining movement stability and control.[17, 19] The minimal differences in GMAX activity observed between individuals with CAI and the control group likely stem from shared neuromuscular recruitment patterns and ingrained habitual behaviors in human motor control mechanisms.[3, 15, 16] Further exploration into nuanced aspects of GMAX involvement in specific activities may provide additional insights into the intricate interplay between anatomy, neuromuscular control, and functional movements.[42, 43, 46]

The meta-analysis results for the GMAX in CAI exhibit heterogeneity to a greater or lesser extent, potentially attributable to the issues mentioned earlier regarding CAI inclusion criteria. At present, the CAI inclusion criteria are commonly based on recommendations by the International Ankle Consortium.[34] However, this set of criteria is now a decade old, and given the evolving landscape of clinical medicine, a reevaluation of CAI diagnostic criteria may be necessary.

Relations between GMED and GMAX

Based on the data of relations (Table II), a weak correlation between the FARs of these two muscles in CAI can be observed, and there are notable differences in the thickness variations of the GMED and GMAX in individuals with CAI during physical activities, indicating significant differences in their recruitment. Conversely, the control group, including both healthy individuals and copers, demonstrates a higher correlation between GMED and GMAX, with similar thickness variation trends. This indicates that the synergistic action between GMED and GMAX should be emphasized, and strengthening GMED may be pivotal in the rehabilitation and management of CAI.

The coordinated synergy between the GMED and GMAX is essential for maintaining pelvic stability, controlling hip movements, and ensuring overall lower limb functionality.[17, 19] These muscles work in tandem during various activities, providing stability during gait, balance, and dynamic movements.[7, 41] In a healthy musculoskeletal system, the GMED and GMAX collaborate seamlessly to stabilize the pelvis and control femoral movements.[17] The GMED, situated on the lateral aspect of the pelvis, contributes to hip abduction and stabilization during single-leg stance, while the more powerful GMAX, spanning the buttocks, plays a pivotal role in hip extension and contributes to overall pelvic stability.[10, 11] Specifically, in individuals with CAI, deficiencies in the coordinated interplay be-

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tween GMED and GMAX may be observed. This lack of synergy can compromise pelvic stability, leading to altered biomechanics during weight-bearing activities. Addressing these deficiencies may be crucial in the rehabilitation process for individuals with CAI.

Strengthening the GMED and its recruitment in individuals with CAI holds particular importance as this muscle contributes to improved hip control, especially during weight-bearing activities.[10, 19] A robust GMED not only aids in pelvic stability but also enhances dynamic balance and postural control.[19] Emphasis on GMED strength is pivotal for individuals with CAI, as it helps address specific deficits in the coordinated functioning of the gluteal muscles.[10, 17] Furthermore, reinforcing the GMED offers additional benefits, such as reducing the risk of lower limb injuries, enhancing overall athletic performance, and promoting optimal musculoskeletal alignment.[10] By prioritizing the development of GMED recruitment in individuals with CAI, clinicians and rehabilitation specialists can effectively target and ameliorate the specific deficits in gluteal muscle coordination, ultimately contributing to improved functional outcomes and enhanced movement control.

When individuals need to maintain balance and enhance postural control, those who have excellent recruitment capacity over their GMED and GMAX benefit greatly.[47] Individuals with ankle dysfunction would require more pelvic stability and hip joint control to assist in posture control, for which adequate GMED and GMAX are necessary.[4, 20, 48]

Additional considerations

The contractile thickness of GMED and GMAX during physical activities in CAI should be subjected to further investigation, since the hypothesis regarding differences in thickness is insufficiently supported by the available data. There are currently few studies on the contractile thickness of GMED and/or GMAX in individuals with CAI; through systematic retrieval, only four relevant studies were found, which just involved four functional movements. Future research efforts should prioritize and expand on this.

Improving the stability of CAI involves considerations beyond strengthening the recruitment capacity of GMED and GMAX. There may be other factors that require attention in enhancing the overall stability of individuals with CAI. One reason why individuals have hip instability may be due to the pain and psychosocial components to CAI.[49] Pain, a common symptom in CAI,[50] could be associated with adaptation within and between muscles in the musculoskeletal system,[51] and kinesiophobia and fear avoidance behavior could have larger and stronger associations with changes in hip kinematics.[52] For instance, if a person walks with a hip drop, it is possibly a pain-avoidance behavior rather than being due to having poor GM muscle activity or a small GM muscle.[49, 52]

Moreover, strategies for controlling stability vary from person to person.[53] Specifically, postural control is commonly seen as having two strategies; the ankle strategy and the hip strategy.[54] An individual employing the ankle strategy would be inclined to use more of the ankle and less of the hip muscles in postural control and functional movements.[44, 55] Based on this, if individuals with CAI tend to use the ankle more in postural control, the influence of the hip muscles on stability may not be obvious.[14, 56] For those individuals with CAI who use the hip strategy, the activity of hip muscles may be more significant.[13, 57] When the tendency of the stability strategy is not clear, there would be confusion regarding the effects of hip muscles on postural control. It would be of value to clearly identify the ankle or hip strategy in individuals with CAI, so as to carry out a targeted intervention.

Nevertheless, strong GMED and GMAX are beneficial for pelvic control and lower limb stability even if the individuals have a tendency to use the ankle strategy.[5, 21, 55] Studies conducting analysis of the contractile thickness of GMED and GMAX by ultrasound, observing muscle size during posture control, would provide useful information for clinical reasoning about abnormal movement control in individuals with CAI.[3, 15, 16]

Limitations of the study

The evidence from this review may be inconclusive because only four studies were able to be selected for performing a meta-analysis and exploring the effects of size of the GMs in individuals with CAI. The small sample compromises the persuasiveness of the study, raising concerns about the robustness of the findings. Additionally, the participants of the included studies were not homogeneous; for instance, there were differences in demographic data such as height, weight, sex, and age. There were also differences in the test limb of controls in the included trials, such as the control limbs that were measured in random order, and matched as the same side to affected or unaffected CAI limbs. Finally, the use of graphical data for some FARs measurements may introduce potential inaccuracies, impacting the precision of the results, and the derived metrics, such as Pearson's Correlation Coefficient and *t*-test (P value and Cohen's-D), may lack precision due to the limited accuracy of the FARs values, particularly when obtained from graphical representations; moreover, these values calculated from a small sample size may lack

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sufficient representativeness and may only serve as a trend indication rather than supporting a definitive conclusion.

Conclusions

Research on the contractile thickness of GMED and GMAX in individuals with CAI is currently limited, and there is a need for increased attention to this topic in future studies. Based on the limited evidence available, it appears that the recruitment capacity of the GMED in individuals with CAI is weaker compared to healthy individuals and Copers. Although the heterogeneity of this result is significant, dividing subgroups according to motor tasks achieved high homogeneity. However, dividing subgroups according to control groups of healthy individuals and Copers still showed high heterogeneity, suggesting that the distinction between Copers and healthy individuals needs further clarification. Additionally, the GMAX in individuals with CAI seems to exhibit relatively normal activity with very small differences compared to control groups. The heterogeneity of this result remains high, and dividing subgroups by motor tasks did not significantly improve homogeneity, indicating the need for further investigation. Lastly, the observed uncoordinated interaction between GMED and GMAX in CAI underscores the importance of addressing these issues. The activity of GMED and GMAX in CAI shows significant differences, while the coordination between these muscles is more pronounced in healthy individuals and Copers. The implication of the findings here is that in the rehabilitation management of CAI, clinicians should prioritize strengthening the recruitment capacity of the GMED to improve postural stability and function.

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Conflicts of interest

The authors certify that there is no conflict of interest with any financial organization regarding the material discussed in the manuscript.

Authors' contributions

Author Luan: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing-original draft preparation, writing-review and editing, visualization; Jeremy Witchalls, Charlotte Ganderton, Roger Adams and Doa El-Ansary: conceptualization, methodology, validation, formal analysis, writing-review and editing; Jia Han: conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing-review and editing, supervision, project administration. All authors read and approved the final version of the manuscript.

History

Article first published online: September 25, 2024. - Manuscript accepted: June 18, 2024. - Manuscript revised: June 13, 2024. - Manuscript received: April 10, 2024.

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SUPPLEMENTARY DIGITAL MATERIAL 2

Supplementary Table II.—The contractile thickness of gluteal muscles during various tasks.[3, 5, 15, 16, 29, 30]

Study	Ultrasound instrumentation	Test and placement	Data acquisition and processing	Summary gluteus medius results of CAI versus Control	Summary gluteus maximus results of CAI versus Control
Koldenhoven[3]	An ultrasound unit (model Acuson Freestyle; Siemens, Mountain View, CA, USA)	Used an 8-MHz wireless transducer to visualize the gluteal muscles. Obtained images with participants in side-lying position for tabletop scans and standing for lateral band walking exercises. For standing and band walking images, secured the transducer in a foam block on the participant's hip using an elastic band, positioning it to visualize both gluteal muscles in the same image.	Images were measured using ImageJ (version 1.50i; National Institutes of Health, Bethesda, MD). In each condition, 3 images were obtained. The average of the 3 measurements was used in the calculations for ARs in each position. Muscle-thickness measures (cm) were obtained by measuring from the inferior aspect of the superior border of the muscle to the superior aspect of the inferior border of the muscle at the widest portion. CSA was determined by tracing the interior aspect of the muscle border to calculate the area (cm^2).	● The GMED activation increased from the non-resisted to the resisted condition in the healthy and coper groups; however, activation in the CAI group did not change between conditions; ● The CAI group had a lower GMED AR with resistance compared with healthy and coper groups; ● All groups demonstrated an increased GMED AR when the band was moved from the lower leg to the forefoot position.	● The GMAX activation increased from the non-resisted to the resisted condition in all groups; ● The CAI group had a higher GMAX AR with resistance compared with healthy and coper groups; ● Similarly, GMAX ARs increased when the band was moved from the lower leg to the forefoot.
DeJong[5]	A portable Siemens ACUSON Freestyle Ultrasound System (Siemens Medical Inc., Mountain View, CA,	Used an 8 MHz wireless transducer with a custom Velcro belt and foam block to	Ultrasound images, measured with ImageJ 1.50i software (National Institutes of Health, USA), were assessed by	● Comparing to healthy groups, the CAI affected limbs had decreased GMED FAR measures from 0 to 40% of gait with moderate to large effect sizes;	● There were no significant differences in GMAX activity at any point in the gait cycle when comparing CAI and

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	USA)	capture B-mode images and walking videos of the gluteal muscles. Positioned the transducer with gel between the greater trochanter and posterior superior iliac spine on participants' right hip.	a blinded investigator. The thickness of gluteal muscles was measured from superior to inferior fascial border in the thickest muscle part. Average thickness for each 10% increment of five gait cycles per limb was calculated. FARs normalized thicknesses during gait to quiet stance.	● Similar outcomes were found for the CAI unaffected limb compared to the healthy group; ● There were no significant differences identified between limbs within healthy or CAI group; ● The CAI group presented with GMED thickness values below quiet standing levels throughout the entire gait cycle.	healthy participants' affected and unaffected limbs, nor between limbs within the CAI and healthy groups; ● GMAX activity was increased above quiet standing measures from terminal stance to mid-swing for all group and limb outcomes.
DeJong[15]	A portable Siemens ACUSON Freestyle Ultrasound System (Siemens Medical Inc., Mountain View, CA, USA)	Used an 8 MHz wireless transducer with a custom Velcro belt and foam block to capture B-mode images and walking videos of the gluteus maximus. Positioned the transducer with gel on participants' right hip, oriented between the greater trochanter and posterior superior iliac spine.	Adjusted USI penetration depth for clear visualization of superior and inferior gluteal muscles fascial borders during walking. Three static standing images were captured as a reference. Image measurements were done using ImageJ 1.50i software. gluteal muscles thickness measured from inferior of superior fascial border to superior of inferior fascial border. Averaged measures at 10% increments for participants at different walking speeds. Normalized walking thickness to quiet standing for FAR	● The coper group presented with significantly increased GMED mean FARs across the entire gait cycle when compared to the CAI outcomes; ● When examining the extent of GMED muscle thickness changes from quiet stance, the CAI group presented with GMED FARs that remained below 1 across the entire gait cycle, indicating that muscle thickness during gait was less than resting thickness measures; ● The coper group presented with FARs above resting values from 10 to 80% of the gait cycle, or from early stance to late swing; ● There were no significant differences in GMED FARs across the three walking speeds for either group; ● The largest effects were noted during the stance phases for each walking speed.	● There were no significant differences for GMAX outcomes when comparing CAI and coper groups; ● GMAX mean FARs remained above resting thickness (FARs>1.0) from approximately 10–50% of gait, or from early to late stance; however, measures remained below resting thickness during swing phases for both groups; ● There were no significant differences for GMAX outcomes when comparing preferred, fast, or standard walking speeds; ● There was not a significant interaction among groups or speeds

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			calculation.		for GMAX measures.
DeJong[16]	A portable ultrasound system (model ACU\SON Freestyle; Siemens Medical Inc, Mountain View, CA, USA)	Used an 8-MHz wireless transducer with a custom belt for ultrasound imaging of gluteal muscles. Created a custom YBT with 3 tape measures at 45° angles for reach directions from a reference horizontal position.	Processed ultrasound images with ImageJ software (version 1.52a; National Institutes of Health, Bethesda, MD). Blinded investigator measured gluteal muscle thickness from inferior to superior fascial border. Calculated average YBT reach distances for anterior, posteromedial, and posterolateral directions, normalized to limb length. Averaged gluteal muscle thickness for each position, normalized to quiet-standing measures for FAR determination.	● No group differences were found for the GMED FARs in any reach direction; ● In the anterior-reach direction, both groups demonstrated thickness measures greater than the quiet stance measures; ● In the posteromedial and posterolateral directions, both groups displayed smaller values than the quiet-stance measures.	● A group difference for the GMAX FAR in the anterior direction was noted, and the CAI group had greater GMAX FAR than the control group; ● However, these differences were not accompanied by differences in the preferential-activation ratios between groups.
DeJong[29]	A Siemen's ACUSON Freestyle Ultrasound System (Siemens Medical Inc., Mountain View, CA, USA).	Used a wireless 8 MHz transducer unilaterally secured on participants' hips with a custom Velcro belt and foam block. Positioned ultrasound head obliquely between greater trochanter and posterior superior iliac spine	This data collection was part of a larger study. USI data processing followed the same procedures that have been described in previous studies.	● Weak correlation between GMED thickness and sEMG amplitudes at a 143-ms lag, with sEMG preceding muscle thickness changes. ● Strong associations observed between GMED thickness and frontal/transverse plane kinetics at 110-ms time frames, with muscle thickness preceding joint moments. ● Most robust associations found with hip abduction/internal rotation kinematics at 99-ms and	● GMAX thickness showed a negative association with sagittal kinetics at a 200-ms lag, indicating thickness measures followed maximal hip flexion moment. ● This association was weaker and had a longer lag time compared to healthy reference data. ● Thickness changes preceded sagittal

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		to visualize gluteal muscles on USI monitor during walking.		66-ms lags, respectively. ● These relationships were stronger and had shorter lag times compared to healthy reference data. ● The relationship with transverse plane kinematics was inverse to healthy data, indicating a difference in muscle thickness and hip rotation motion for CAI.	kinematics by 220 ms, which was also weaker and at a longer time interval than healthy reference data.
DeJong[30]	A portable Siemens ACUSON Freestyle Ultrasound System (Siemens Medical Inc, Mountain View, CA, USA)	Used an 8-MHz wireless linear transducer to obtain GMAX and GMED image and video clip data. The transducer was secured unilaterally on patients' hips using a custom Velcro belt with foam block.	Each 10-second USI video clip, comprising 150 frames and 9 gait cycles per patient, was analyzed. GMAX and GMED thicknesses were measured using ImageJ, spanning from superior to inferior fascial border. This process was repeated for all quiet standing measures across 55 frames. Average thickness measures over 5 gait cycles were normalized to quiet stance for GMAX and GMED FARs.	● Regardless of group allocation, there was a statistically significant time main effect demonstrating increased GMED FARs across the entire gait cycle compared with baseline measures with large corresponding effect sizes.	● The GMAX measures were not significantly different across time points. ● When comparing the GMAX and GMED FARs between groups at both time points, there were no statistically significant group by time interactions for either muscle across the entire gait cycle.

FAR: Functional Activity Ratio (muscle thickness during activity / muscle thickness quiet); USI: musculoskeletal ultrasound imaging; GMED: gluteus medius; GMAX: gluteus maximus; YBT: Y-Balance Test; CSA: Cross-sectional area; AR: activation ratio; sEMG: surface electromyography.

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SUPPLEMENTARY DIGITAL MATERIAL 1

Supplementary Table I.—Participants' characteristics of included studies.[3, 5, 15, 16, 29, 30]

Study	Participants: Chronic ankle instability			Participants: Control		
Task	Inclusion criteria	Characteristics	Test limb	Inclusion criteria	Characteristics	Test limb
Koldenhoven[3] 1. Hip-abduction (side lying, resistance: none) 2. Hip-abduction (side lying, resistance: manual) 3. Hip-abduction (lateral walking, resistance: elastic band placed in lower leg) 4. Hip-abduction (lateral walking, resistance: elastic band placed in forefoot)	Participants had to have incurred at least 1 significant LAS at least 12 months before participation, had not sprained their ankle within 8 weeks of participation, experienced sensations of giving way or residual perceived instability, and scored >10 on the IdFAI, <90% on the FAAM-ADL, and <85% on the FAAM-sport.	N.=19 Age (years) 20.9 (4.8) Height (m) 1.68 (0.09) Weight (kg) 70.8 (14.6) Sex (F/M) 14/5	CAI involved or worse limb in the case of bilateral CAI	①: Healthy participants had no history of ankle sprain. ②: Copers were defined as having experienced 1 significant LAS at least 12 months before study participation and no episodes of giving way or perceived instability (IdFAI≤10, FAAM-ADL≥99% and FAAM-sport≥97%).	①: N.=16 Age (years) 19.8 (0.8) Height (m) 1.73 (0.11) Weight (kg) 69.6 (15.2) Sex (F/M) 8/8 ②: N=17 Age (years) 20.4 (2.1) Height (m) 1.71 (0.09) Weight (kg) 68.9 (9.6) Sex (F/M) 11/6	Limbs were assessed in random order
DeJong[5] Walking Speed: 1.35 m/s	At least one significant ankle sprain 12 months or more prior to enrollment, no current ankle injuries ≤3 months of the study, <90% on the FAAM-ADL and <85% on the FAAM-sport, ≥11 on the IdFAI, and<24 on the CAIT.	N=20 Age (years) 21.7 (2.32) Height (m) 1.73 (0.11) Weight (kg) 74.26 (15.24) Sex (F/M) 10/10	Affected limb (CAI involved or worse limb in the case of bilateral CAI) and unaffected limb (the contralateral limb)	Healthy participants without a history of ankle sprains	N=20 Age (years) 21.2 (2.79) Height (m) 1.73 (0.15) Weight (kg) 70.89 (12.18) Sex (F/M) 10/10	Matched as the same side to affected and unaffected CAI limbs based on participant sex and height
DeJong[15] 1. Walking Speed: 1.35 m/s	Participants have at least one significant ankle sprain sustained 12 months or more prior to study enrollment with residual	N=14 Age (years) 21 (3) Height (m)	Ankle sprain limb	Copers were classified as having one significant ankle sprain at least 12 months prior to study	N=14 Age (years) 21 (2) Height (m)	Ankle sprain limb

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2. Walking Speed: preferred 3. Walking Speed: 120% of preferred (fast)	disability, classified as ≤90 on the FAAM-ADL, and ≤85 on the FAAM-sport.	1.68 (0.08) Weight (kg) 68.9 (14.4) Sex (F/M) 14/0		collection without any residual disability, determined as ≥99 on the FAAM-ADL, and ≥97 on the FAAM-Sport.	1.67 (0.04) Weight (kg) 64.3 (7.0) Sex (F/M) 14/0	
DeJong[16] 1. Balance Anterior 2. Balance Posteromedial 3. Balance Posterolateral	Participants had sustained at least 1 substantial ankle sprain 12 months or more before enrollment and no ankle injuries within 3 months of the study, and they had to score >11 on the IdFAI, <90% on the FAAM-ADL, <80% on the FAAM-sport, and <24 on the CAIT.	N=20 Age (years) 21.7 (2.32) Height (m) 1.73 (0.11) Weight (kg) 74.26 (15.24) Sex (F/M) 10/10	CAI involved or worse limb in the case of bilateral CAI	Healthy participants had no history of ankle sprain.	N=20 Age (years) 21.2 (2.79) Height (m) 1.73 (0.15) Weight (kg) 70.89 (12.18) Sex (F/M) 10/10	Matched to the limbs of the CAI-group participants based on sex and height
DeJong[29] Walking Speed: 1.35 m/s	Participants were required to have a history of at least 1 significant lateral ankle sprain at least 12 months prior to study participation, self-reported dysfunction (FAAM-sport ≤85%) and feelings of instability (IdFAI ≥11).	N=15 Sex (F/M) 15/0	CAI involved or worse limb in the case of bilateral CAI	-	-	-
DeJong[30] Walking Speed: 1.34 m/s	At least one significant ankle sprain no less than 12 months prior to study enrollment (≤90 on the FAAM-ADL). Participants were also required to score ≤85 on the FAAM-sport subscale.	N=9 Age (years) 23 (5) Height (m) 1.686 (0.08) Weight (kg) 71.95 (15.12) Sex (F/M) 7/2	CAI worst scoring limb as reported on ankle-specific questionnaires	Same as another group	N=9 Age (years) 21 (4) Height (m) 1.74 (0.115) Weight (kg) 71.23 (13.16) Sex (F/M) 7/2	Same as another group

FAAM-ADL: Foot and Ankle Ability Measure Activities of Daily Living Subscale; FAAM-sport: Foot and Ankle Ability Measure Sport Subscale; IdFAI: Identification of Functional Instability scale; CAIT: Cumberland Ankle Instability Tool; LAS: lateral ankle sprain.